

A restart is not a recovery.

Voice-led investigation with cited observations, explicit approval, and measured recovery checks.

- Built for the AssemblyAI Voice Agent Hackathon
- Find evidence → approve a change → verify the incident

Keep the investigation moving.

During an outage, engineers move between conversations, logs, service health, and follow-up notes.

- A successful command is not proof of a recovered service.
- Engineers need to inspect the diagnosis, control the action, and verify its effect.
- Target users: on-call engineers and small platform teams.

From a question to a reviewed action.

Ask about the incident, inspect evidence, and decide what should happen next.

- Investigate → capture health and logs, then inspect evidence-linked hypotheses.
- Stage → show the exact action, service, parameters, and approval expiry.
- Approve or cancel → execute once, then show the observed result.
- Verify → compare before/after health and check the remaining incident impact.

Two voice paths. One tool boundary.

Both voice engines share the same session and approval logic.

- Managed: AssemblyAI Voice Agent API with 24 kHz PCM audio and client tools.
- Custom: AssemblyAI Streaming v3 STT → configured LLM or labeled scripted commands → speech output.
- Investigation briefs and reports: AssemblyAI LLM Gateway analyzes captured evidence.
- Provider failures and local fallbacks are visible in the workspace.

Approval is part of execution.

A model request stages a mutation. A second tool call does not authorize it.

- Each action has a unique ID, exact target, and a 30-second approval window.
- Cancellation, expiry, role checks, and replay rejection are enforced in the backend.
- Live infrastructure requires an operator token.
- Automatic approval is restricted to explicitly enabled simulation.

Open the observation behind the hypothesis.

IncidentVoice
VOICE INCIDENT WORKSPACE

STANDBY STT: - Total: - Settings

OPERATIONS / INCIDENT WORKSPACE

Less typing. Faster triage.
Investigate together. Make the next move with confidence.

Investigate Incident Create Incident report

SEV-1 INC-8942
Payment Gateway Outage (HTTP 503 Spike)
Investigating / Demo simulation · no live changes

NEEDS ATTENTION
4
2 active critical services

CLUSTER HEALTH
1 / 5
Healthy in this scenario

Voice SRE Incident Copilot
Speak naturally. Stay in control. Copilot Standby

Operator 02:08:51 PM
Investigate the incident

IncidentVoice SRE 02:08:54 PM
4 services need investigation. 17 observations captured; root cause remains unverified. AssemblyAI proposed 3 hypotheses to test. Review the cited evidence before approving a change.

QUICK Investigate the incident What alerts are firing? Inspect payment-service Show dependency topology

Start voice MIC STANDBY

Type SRE command or speak into mic...

Enter to send Changes require your approval

INVESTIGATE
The bigger picture 5 services

Brief Services Runbooks Topology Demo lab

Investigate → Review & approve → Verify

simulation evidence Captured 02:08 PM Refresh

ASSEMBLYAI INVESTIGATION
What the evidence suggests

4 services need investigation. 17 observations captured; root cause remains unverified. AssemblyAI proposed 3 hypotheses to test.

17 observations · 5 services

01 HYPOTHESIS · UNVERIFIED
Database connection exhaustion and lock contention

The database has hit its maximum connection limit, causing the payment-service to exhaust its connection pool and crash due to memory pressure while waiting for locks.

Supporting observations E01 ↓ E02 ↓ E03 ↓ E04 ↓ E09 ↓

Next check
Inspect the latest logs for order-db and compare them with the cited snapshot before choosing a

Captured facts.
Inspectable hypotheses.

- 17 observations from five simulated services in this demonstration.
- AssemblyAI analysis linked to exact evidence IDs.
- Read-only next checks; stale evidence is flagged.

A healthy target. An incident still in progress.

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Speak naturally. Stay in control. Copilot Standby

IncidentVoice SRE 02:08:55 PM
Simulation applied. Graceful rolling restart executed for payment-service. Clean pods spawned.

Operator 02:08:55 PM
Verify recovery

IncidentVoice SRE 02:08:55 PM
Recovery is incomplete: ingress-gateway, order-db, redis-cache still need attention or health verification.

QUICK Investigate the incident What alerts are firing? Inspect payment-service Show dependency topology

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AFTER THE ACTION simulation

The target is healthy.

payment-service - restart pod

Metric	Value
Status	critical → healthy
Error rate	42.6% → 0.5%
P99 latency	2850 ms → 65 ms

Simulated result. This check covers the target service; other services may still be affected.

Check the whole cluster

✓ Recovery is still in progress
Recovery is incomplete: ingress-gateway, order-db, redis-cache still need attention or health verification.

simulation evidence Captured 02:08 PM Refresh

Service state has changed since this brief. Refresh before relying on its hypotheses.

Success needs context.

- Approve the exact service restart.
- Compare observed before/after target health.
- Check the whole cluster: three services still need attention here.
- Export the evidence and outcomes for the next engineer.

Sell the workflow, measure the outcome.

Start with small platform teams that own on-call incidents.

- Buyer: platform engineering or SRE lead; daily user: on-call responder.
- Pilot: compare time to first cited diagnosis, approval correctness, and verified recovery against the team's existing workflow.
- Business hypothesis: per-team subscription with voice usage charged separately; pricing and demand are unvalidated.
- Next: deploy an HTTPS demo, test a real microphone session, and recruit pilot operators. Live actions remain limited.